

PURUSOTHKUMAR S

Data Engineer

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SUMMARY

Data Engineer with 3+ years of experience designing and developing scalable batch and real-time data pipelines, CDC workflows, and cloud-based lakehouse architectures. Hands-on experience with CDC ingestion (Kafka, Flink, Debezium), Apache Iceberg and Delta Lake table formats, medallion architecture, and dimensional data modeling. Skilled in Python, SQL, Airflow, Databricks, Snowflake, Trino, and Apache Doris — processing 5M+ records daily, cutting data delivery time by 60%+, and improving pipeline efficiency through optimization and fault-tolerant streaming design.

CORE SKILLS

Languages & Scripting: Python, SQL, PySpark, Pandas, NumPy, Bash/Linux scripting

Cloud Platform: AWS (S3, EC2, Redshift)

Streaming & Big Data: Apache Kafka, Apache Flink, Apache Spark, PySpark, Apache Airflow, Databricks, Delta Lake, Apache Iceberg, CDC, Snowflake

CI/CD & Version Control: Git, GitHub, GitHub Actions-style workflows, Jenkins

Core Competencies: ETL/ELT, Data Warehousing, Lakehouse Architecture, Data Migration, Data Modeling, Real-Time Processing, Data Quality, Event-Driven Architecture, Medallion Architecture

BI & Visualization: Power BI

Tools: VS Code, Postman, DBeaver, PGAdmin, Trino

PROFESSIONAL EXPERIENCE

Data Engineer — Span Technology Services Pvt Ltd

Oct 2023 – Present

- Designed and maintained scalable, distributed batch and real-time data pipelines (Python, SQL, Apache Kafka, Apache Flink, PostgreSQL), processing **5M+ records per day** across production workloads **with zero data loss**.
- Built and maintained **CDC** pipelines using **Debezium** connectors to capture changes from SQL Server, PostgreSQL, and MySQL and publish change events into Kafka for downstream real-time processing.
- Architected a real-time payment reconciliation pipeline (Kafka + Flink stateful processing, exactly-once semantics) with automated mismatch detection and Slack alerting — cutting manual **reconciliation effort by 85%** and **duplicate payments by 60%**.
- Implemented **Apache Iceberg**-based **Bronze/Silver/Gold** lakehouse architecture for scalable storage, data transformation, schema evolution, and analytical workloads.
- Designed and implemented OLAP data models using **Fact and Dimension tables** for analytical and reporting workloads, enabling efficient querying and downstream BI consumption.
- Cataloged and governed data assets using **OpenMetadata**, improving data discoverability, metadata management, lineage tracking, and cross-team visibility into pipeline dependencies.
- Built and orchestrated ETL/ELT workflows using **Apache Airflow**, reducing end-to-end **data delivery time by 60%** through automated scheduling, dependency management, and parallel task execution.
- Reduced **pipeline failures by 90%** and improved recovery time by implementing checkpointing and savepointing in Flink, ensuring fault-tolerant, reliable streaming pipelines.
- Resolved **20+ critical production** pipeline incidents, performing root-cause analysis, bug fixes, and performance tuning to restore SLA-compliant data delivery within tight timelines.
- Optimized PySpark transformations, SQL queries, and data processing pipelines, improving overall pipeline efficiency by 50%+ and reducing manual intervention by over 30%.
- Collaborated with data scientists, analytics, and product teams to deliver accurate, timely datasets powering business reporting and data-driven decisions.
- Worked within CI/CD workflows using Git/GitHub and Jenkins for reliable deployment and engineering standards documentation.

Software Engineering Trainee — Decision Minds Pvt Ltd

Jan 2023 – Sep 2023

- Built scalable ETL pipelines in Python supporting incremental data loading, transformation, and warehouse materialization in **Snowflake**.
- Migrated data from PostgreSQL, Oracle, and AWS S3 into Snowflake, implementing profiling, validation, transformation, and reconciliation to ensure data integrity and quality.
- Automated data ingestion, transformation, and loading into Snowflake, reducing manual **data-processing effort by 50%**.
- Built interactive **Power BI dashboards** to visualize ETL results, validate data, and deliver business insights to stakeholders.

PROJECTS

Retail Data Lakehouse — Databricks, PySpark, Delta Lake

- Built an end-to-end retail lakehouse in Databricks using PySpark and Spark SQL, processing customer, product, order, and sales data through Bronze/Silver/Gold **medallion architecture**.
- Used Delta Lake ACID transactions, schema enforcement/evolution, and time travel for reliable, auditable data management; implemented Unity Catalog for centralized data governance and access control.
- Optimized Spark transformations with partitioning and efficient DataFrame operations to improve processing performance.

Product Data Platform — TaxBandits & TruckLogics

- Maintained legacy pipeline (SQL → Kafka → Flink → PostgreSQL) processing **2M+ records daily**, resolving recurring late data arrivals, null occurrences, and dashboard slowness caused by PostgreSQL bottlenecks.
- Diagnosed root causes including Kafka consumer lag, Flink misconfigurations, and unindexed tables — **reducing data delays by 40%** and improving pipeline stability.
- Migrated to a modernized architecture (SQL → Kafka → MinIO Schema Registry → Flink → Apache Doris), eliminating late-arrival and null-value issues through improved checkpointing and schema validation.
- Consolidated MongoDB and SQL sources into Apache Doris, replacing slow multi-join queries and improving dashboard performance by ~70%; exposed REST API endpoints for low-latency dashboard consumption.

CERTIFICATIONS & AWARDS

- Microsoft Certified: Fabric Data Engineer Associate (DP-700) — Microsoft, July 2026
- Super Sparkler Award, Span Technology Services Pvt Ltd — recognized for delivering critical data engineering projects on tight timelines and cross-functional collaboration

EDUCATION

B.E., Electronics and Communication Engineering — Anna University, Chennai

Aug 2018 – Jul 2022

Sri Balaji Chocklingam Engineering College | CGPA: 8.02/10.0